0.1 Construction of the Real Numbers: Completeness Axiom, Dedekind Cuts, and Cauchy Sequences

The fundamental inadequacy of the rational number field Q lies in its lack of completeness; it contains gaps, most famously demonstrated by the absence of a rational root for the equation $x^2 = 2$. To establish a rigorous foundation for analysis, we must construct the field of real numbers R such that it satisfies the Least Upper Bound property. We examine two classical constructions: Dedekind cuts and equivalence classes of Cauchy sequences, ultimately proving their equivalence in yielding a complete ordered field.

0.1.1 Dedekind Cuts

A **Dedekind cut** is a subset $\alpha \subset Q$ satisfying the following three properties:

1. **Nontriviality:** $\alpha \neq \emptyset$ and $\alpha \neq Q$.

2. **Downward Closure:** If $p \in \alpha$, $q \in Q$, and $q < p$, then $q \in \alpha$.

3. **No Greatest Element:** If $p \in \alpha$, there exists $r \in \alpha$ such that $p < r$.

Let R denote the set of all Dedekind cuts. We define an ordering on R by strict set inclusion: $\alpha < \beta$ if and only if $\alpha\beta$.

[Completeness Axiom for Cuts] The set R of Dedekind cuts satisfies the Least Upper Bound property. That is, every non-empty subset $A \subset R$ that is bounded above possesses a supremum in R.

Let A be a non-empty subset of R that is bounded above. This implies there exists some cut $\beta \in R$ such that $\alpha \subset \beta$ for all $\alpha \in A$. We define the candidate supremum γ as the union of all cuts in A:

$$\gamma = \bigcup_{\alpha \in A} \alpha$$

We must first verify that γ is a valid Dedekind cut.

1. Since A is non-empty, there exists some $\alpha_0 \in A$. Because α_0 is a cut, $\alpha_0 \neq \emptyset$, hence $\gamma \neq \emptyset$. Furthermore, since every $\alpha \in A$ is bounded above by β, $\gamma \subset \beta$. As $\beta \neq Q$, it follows that $\gamma \neq Q$.

2. Let $p \in \gamma$ and $q \in Q$ with $q < p$. By definition of γ, $p \in \alpha$ for some $\alpha \in A$. Since α is downward closed, $q \in \alpha$, which implies $q \in \gamma$.

3. Let $p \in \gamma$. Then $p \in \alpha$ for some $\alpha \in A$. Since α has no greatest element, there exists $r \in \alpha$ such that $p < r$. Thus, $r \in \gamma$, demonstrating that γ has no greatest element.

Therefore, $\gamma \in R$. By construction, for any $\alpha \in A$, $\alpha \subset \gamma$, so γ is an upper bound for A. Suppose $\delta < \gamma$ is another upper bound for A. Then there exists $p \in \gamma$ such that $p \notin \delta$. Since $p \in \gamma$, $p \in \alpha$ for some $\alpha \in A$. But this implies $\alpha \not\subset \delta$, contradicting the assumption that δ is an upper bound for A. Thus, γ is the least upper bound of A.

0.1.2 Cauchy Sequences and the Completion of Q

An alternative, topologically motivated construction of R relies on Cauchy sequences.

A sequence $(q_n)_{n=1}^{\infty}$ in Q is a **Cauchy sequence** if for every rational $\epsilon > 0$, there exists an integer $N \in N$ such that for all $m, n \geq N$,

$$|q_m - q_n| < \epsilon$$

Let $\mathcal{C}$ denote the set of all Cauchy sequences in Q. We define an equivalence relation $\sim$ on $\mathcal{C}$ by declaring $(q_n) \sim (p_n)$ if and only if $\lim_{n\to\infty} |q_n - p_n| = 0$.

The relation $\sim$ is an equivalence relation on $\mathcal{C}$.

Reflexivity and symmetry are immediate from the properties of the absolute value function. For transitivity, suppose $(q_n) \sim (p_n)$ and $(p_n) \sim (r_n)$. Let $\epsilon > 0$ be given. There exist $N_1, N_2 \in N$ such that $|q_n - p_n| < \frac{\epsilon}{2}$ for all $n \geq N_1$ and $|p_n - r_n| < \frac{\epsilon}{2}$ for all $n \geq N_2$. Let $N = \max(N_1, N_2)$. For $n \geq N$, the triangle inequality yields:

$$|q_n - r_n| \leq |q_n - p_n| + |p_n - r_n| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Hence, $\lim_{n\to\infty} |q_n - r_n| = 0$, proving $(q_n) \sim (r_n)$.

We construct R as the quotient space $\mathcal{C}/\sim$. An element $x \in R$ is thus an equivalence class of Cauchy sequences of rational numbers.

The field R constructed via Cauchy sequences is complete. That is, every Cauchy sequence in R converges to a limit in R.

Let $(x_n)_{n=1}^{\infty}$ be a Cauchy sequence in R. Each x_n is an equivalence class of rational Cauchy sequences. For each $n \in N$, choose a representative rational sequence $(q_{n,k})_{k=1}^{\infty} \in x_n$. Since x_n represents a real number, there exists $K_n \in N$ such that for all $j, k \geq K_n$,

$$|q_{n,j} - q_{n,k}| < \frac{1}{n}$$

We extract a diagonal sequence $y_n = q_{n,K_n}$. We claim that $(y_n)_{n=1}^{\infty}$ is a Cauchy sequence in Q, and its equivalence class $y \in R$ is the limit of the sequence (x_n).

Let $\epsilon > 0$. Choose $N \in N$ such that $\frac{3}{N} < \epsilon$. Since (x_n) is Cauchy in R, there exists $M \geq N$ such that for all $m, n \geq M$, $|x_m - x_n| < \frac{\epsilon}{3}$. By the definition of the real metric, this implies that for sufficiently large k, $|q_{m,k} - q_{n,k}| < \frac{\epsilon}{3}$.

Consider $m, n \geq M$. We have:

$$|y_m - y_n| = |q_{m,K_m} - q_{n,K_n}| \leq |q_{m,K_m} - q_{m,k}| + |q_{m,k} - q_{n,k}| + |q_{n,k} - q_{n,K_n}|$$

For $k \geq \max(K_m, K_n)$ large enough to satisfy the inter-sequence bound, this yields:

$$|y_m - y_n| < \frac{1}{m} + \frac{\epsilon}{3} + \frac{1}{n} \leq \frac{1}{M} + \frac{\epsilon}{3} + \frac{1}{M} \leq \frac{2}{N} + \frac{\epsilon}{3} < \epsilon$$

Thus, (y_n) is a Cauchy sequence in Q. Let $y = [(y_n)]$. A similar bounding argument demonstrates that $|x_n - y| \to 0$ as $n \to \infty$, proving that R is complete under the metric induced by this construction.

0.1 Topology of R: Open/Closed Sets, Compact Sets, and the Heine-Borel Theorem

Having constructed the real numbers as a complete ordered field, we now formalize its topological structure. The standard topology on R is induced by the Euclidean metric $d(x,y) = |x-y|$.

Let $x \in R$ and $\epsilon > 0$. The ϵ-neighborhood of x is the open interval $V_\epsilon(x) = (x-\epsilon, x+\epsilon)$. A set $U \subset R$ is **open** if for every $x \in U$, there exists an $\epsilon > 0$ such that $V_\epsilon(x) \subset U$. A set $F \subset R$ is **closed** if its complement $F^c = R \setminus F$ is open.

A point $x \in R$ is a **limit point** of a set $A \subset R$ if every neighborhood $V_\epsilon(x)$ intersects A at a point distinct from x. That is, $(V_\epsilon(x) \setminus \{x\}) \cap A \neq \emptyset$ for all $\epsilon > 0$.

A set $F \subset R$ is closed if and only if it contains all its limit points.

Suppose F is closed, and let x be a limit point of F. We proceed by contradiction. If $x \notin F$, then $x \in F^c$. Because F^c is open by definition, there exists $\epsilon > 0$ such that $V_\epsilon(x) \subset F^c$. This implies $V_\epsilon(x) \cap F = \emptyset$, which contradicts the premise that x is a limit point of F. Therefore, $x \in F$.

Conversely, suppose F contains all its limit points. We must show that F^c is open. Let $y \in F^c$. Since $y \notin F$, y is not a limit point of F. By definition, there exists an $\epsilon > 0$ such that $V_\epsilon(y)$ contains no points of F other than possibly y itself. But $y \notin F$, so $V_\epsilon(y) \cap F = \emptyset$. This means $V_\epsilon(y) \subset F^c$. Since this holds for all $y \in F^c$, F^c is open, and thus F is closed.

An **open cover** for a set $E \subset R$ is a collection of open sets $\{G_\alpha\}_{\alpha \in A}$ such that $E \subset \bigcup_{\alpha \in A} G_\alpha$. A set $K \subset R$ is **compact** if every open cover of K contains a finite subcover.

[Heine-Borel Theorem] A subset $K \subset R$ is compact if and only if it is closed and bounded.

() Assume K is compact. We first show K is bounded. Consider the open cover consisting of intervals $I_n = (-n, n)$ for all $n \in N$. Since $K \subset \bigcup_{n=1}^{\infty} I_n$, compactness guarantees a finite subcover $\{I_{n_1}, I_{n_2}, \ldots, I_{n_k}\}$. Let $N = \max\{n_1, \ldots, n_k\}$. Then $K \subset (-N, N)$, meaning K is bounded.

Next, we show K is closed by proving K^c is open. Let $p \in K^c$. For each $q \in K$, choose disjoint open neighborhoods U_q of p and V_q of q; explicitly, one may take $\epsilon = \frac{|p-q|}{2}$ and let $U_q = V_\epsilon(p)$ and $V_q = V_\epsilon(q)$. The collection $\{V_q\}_{q \in K}$ is an open cover of K. By compactness, there exists a finite subcover $\{V_{q_1}, \ldots, V_{q_m}\}$. Let $W = \bigcap_{i=1}^{m} U_{q_i}$. As a finite intersection of open sets, W is open. Furthermore, $p \in W$, and $W \cap V_{q_i} = \emptyset$ for all $i = 1, \ldots, m$. Because $\{V_{q_i}\}$ covers K, it follows that $W \cap K = \emptyset$. Thus, $W \subset K^c$, proving K^c is open and K is closed.

() Assume K is closed and bounded. Because K is bounded, $K \subset [a,b]$ for some real numbers $a < b$. We first prove $[a,b]$ is compact. Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of $[a,b]$. Define the set S as:

$$S = \{x \in [a,b] \mid theinterval[a,x]iscoveredbyafinitesubcollectionof\{G_\alpha\}\}$$

Since $a \in [a,b]$, there is some G_{α_0} containing a, meaning $[a,a]$ is finitely covered. Thus $a \in S$, and $S \neq \emptyset$. Furthermore, S is bounded above by b. By the Completeness Axiom established in Section 1.1, S possesses a supremum $c = \sup S$, where $a \leq c \leq b$.

Since $c \in [a,b]$, $c \in G_\beta$ for some $\beta \in A$. Because G_β is open, there exists $\epsilon > 0$ such that $(c-\epsilon, c+\epsilon) \subset G_\beta$. By the definition of the supremum, there exists $x \in S$ such that $c - \epsilon < x \leq c$. Because $x \in S$, $[a,x]$ is covered by a finite subcollection $\mathcal{F} \subset \{G_\alpha\}$. The augmented collection $\mathcal{F} \cup \{G_\beta\}$ is finite and covers $[a, c+\frac{\epsilon}{2})$. If $c < b$, this implies $c + \frac{\epsilon}{2} \in S$, contradicting $c = \sup S$. Hence, $c = b$, and by the same logic, $b \in S$. Therefore, $[a,b]$ is finitely covered, making it compact.

Finally, let $\{U_\lambda\}_{\lambda \in \Lambda}$ be an open cover of K. Since K is closed, K^c is open. The collection $\{U_\lambda\}_{\lambda \in \Lambda} \cup \{K^c\}$ forms an open cover of the compact set $[a,b]$. This extended cover admits a finite subcover Φ. Removing K^c from Φ leaves a finite subcollection of $\{U_\lambda\}$ that still fully covers K. Thus, K is compact.

0.1 $\epsilon - \delta$ Limits, Uniform Continuity, and the Bolzano-Weierstrass Theorem

Building upon the topological framework of R, we rigorously define the limiting behavior of functions and sequences, transitioning from local properties to global guarantees on compact domains.

Let $E \subset R$, and let c be a limit point of E. A function $f : E \to R$ has a **limit** $L \in R$ as x approaches c, denoted $\lim_{x \to c} f(x) = L$, if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in E$,

$$0 < |x - c| < \delta |f(x) - L| < \epsilon$$

From this local definition, we define continuity: a function f is continuous at $c \in E$ if $\lim_{x \to c} f(x) = f(c)$. If this holds for all points in E, f is continuous on E. While continuity is a local property (the choice of δ may depend on both ϵ and c), uniform continuity establishes a global bound independent of c.

A function $f : E \to R$ is **uniformly continuous** on E if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in E$,

$$|x - y| < \delta |f(x) - f(y)| < \epsilon$$

[Heine-Cantor Theorem] If $K \subset R$ is compact and $f : K \to R$ is continuous, then f is uniformly continuous on K.

Let $\epsilon > 0$ be given. Because f is continuous on K, for each $p \in K$, there exists a $\delta_p > 0$ such that for all $x \in K$,

$$|x - p| < \delta_p |f(x) - f(p)| < \frac{\epsilon}{2}$$

Consider the collection of open neighborhoods $V_p = (p - \frac{\delta_p}{2}, p + \frac{\delta_p}{2})$ for all $p \in K$. The collection $\{V_p\}_{p \in K}$ forms an open cover of K. Because K is compact, there exists a finite subcover corresponding to points $p_1, p_2, \ldots, p_n \in K$.

Define $\delta = \frac{1}{2} \min\{\delta_{p_1}, \delta_{p_2}, \ldots, \delta_{p_n}\}$. Since the set of radii is finite and each $\delta_{p_i} > 0$, we have $\delta > 0$. Let $x, y \in K$ such that $|x - y| < \delta$. Because the finite collection $\{V_{p_i}\}_{i=1}^n$ covers K, there is some integer $k \in \{1, \ldots, n\}$ such that $x \in V_{p_k}$. This implies:

$$|x - p_k| < \frac{\delta_{p_k}}{2}$$

By the triangle inequality, we bound the distance between y and p_k:

$$|y - p_k| \leq |y - x| + |x - p_k| < \delta + \frac{\delta_{p_k}}{2}$$

Since $\delta \leq \frac{\delta_{p_k}}{2}$, we deduce $|y - p_k| < \delta_{p_k}$. We evaluate the difference in functional values using the continuity of f at p_k. Both x and y are strictly within δ_{p_k} of p_k, yielding:

$$|f(x) - f(y)| \leq |f(x) - f(p_k)| + |f(p_k) - f(y)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus, f is uniformly continuous on K.

We conclude this chapter with a fundamental property of sequences in bounded domains, relying heavily on the completeness axiom.

[Bolzano-Weierstrass Theorem] Every bounded sequence in R possesses a convergent subsequence.

Let $(x_n)_{n=1}^{\infty}$ be a bounded sequence in R. There exists a closed interval $I_0 = [a_0, b_0]$ such that $x_n \in I_0$ for all $n \in N$. We proceed by constructing a nested sequence of closed intervals.

Bisect I_0 into two closed subintervals: $[a_0, \frac{a_0+b_0}{2}]$ and $[\frac{a_0+b_0}{2}, b_0]$. Since I_0 contains infinitely many terms of the sequence (x_n), at least one of these two subintervals must contain infinitely many terms. Let $I_1 = [a_1, b_1]$ denote this subinterval. We select an index n_1 such that $x_{n_1} \in I_1$.

Assume inductively that we have constructed a closed interval $I_k = [a_k, b_k]$ containing infinitely many terms of (x_n), with length $\frac{b_0-a_0}{2^k}$, and we have chosen an index $n_k > n_{k-1}$ such that $x_{n_k} \in I_k$. Bisect I_k into two equal halves. At least one half contains infinitely many terms of (x_n); call it $I_{k+1} = [a_{k+1}, b_{k+1}]$. Choose an index $n_{k+1} > n_k$ such that $x_{n_{k+1}} \in I_{k+1}$.

This process recursively generates a sequence of nested closed intervals $I_0 \supset I_1 \supset I_2 \supset \ldots \supset I_k \supset \ldots$ and a subsequence $(x_{n_k})_{k=1}^{\infty}$ where $x_{n_k} \in I_k$ for all k. The sequences of endpoints (a_k) and (b_k) are monotonic: (a_k) is non-decreasing and bounded above by b_0, while (b_k) is non-increasing and bounded below by a_0. By the Monotone Convergence Theorem (derived from the Least Upper Bound property), both converge. Since the interval length is $b_k - a_k = \frac{b_0-a_0}{2^k}$, we have $\lim_{k\to\infty}(b_k - a_k) = 0$, proving they converge to the identical limit $L \in R$.

Because $a_k \leq x_{n_k} \leq b_k$ for all k, the Squeeze Theorem forces $\lim_{k\to\infty} x_{n_k} = L$. Thus, (x_{n_k}) converges.

1 Proof-Based Calculus & Integration

1.1 The Mean Value Theorem and Rigorous Generalizations

The transition from the topological foundations of R to differential calculus requires establishing the relationship between the local behavior of a function's derivative and its global variations over an interval. The cornerstone of this bridge is the Mean Value Theorem, which relies essentially on the topological compactness of closed intervals.

Before proving the Mean Value Theorem, we first establish Rolle's Theorem, which itself depends on the Extreme Value Theorem (a direct consequence of the Heine-Borel theorem established in Chapter 1).

[Fermat's Theorem on Stationary Points] Let $f : (a, b) \to R$ be a function. If f attains a local maximum or local minimum at a point $c \in (a, b)$, and if f is differentiable at c, then $f'(c) = 0$.

Assume f attains a local maximum at c. By definition, there exists an $\epsilon > 0$ such that for all $x \in (c - \epsilon, c + \epsilon)$, $f(x) \leq f(c)$. For $x \in (c, c + \epsilon)$, $x - c > 0$ and $f(x) - f(c) \leq 0$, yielding a difference quotient $\frac{f(x)-f(c)}{x-c} \leq 0$. Taking the right-hand limit as $x \to c^+$, we find $f'(c) \leq 0$. Conversely, for $x \in (c - \epsilon, c)$, $x - c < 0$ and $f(x) - f(c) \leq 0$, yielding $\frac{f(x)-f(c)}{x-c} \geq 0$. Taking the left-hand limit as $x \to c^-$, we find $f'(c) \geq 0$. Since f is differentiable at c, the left and right limits must coincide, forcing $f'(c) = 0$. The proof for a local minimum proceeds symmetrically.

[Rolle's Theorem] Let $f : [a, b] \to R$ be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b). If $f(a) = f(b)$, then there exists at least one point $c \in (a, b)$ such that $f'(c) = 0$.

Because $[a, b]$ is compact and f is continuous, the Extreme Value Theorem guarantees that f attains its global maximum M and global minimum m on $[a, b]$. If $M = m$, then $f(x)$ is constant on $[a, b]$. Consequently, $f'(x) = 0$ for all $x \in (a, b)$, and any $c \in (a, b)$ satisfies the theorem. If $M > m$, then since $f(a) = f(b)$, f must attain at least one of its extrema at a point c in the open interval (a, b). By Fermat's Theorem, because f is differentiable at this interior extremum, $f'(c) = 0$.

[Lagrange Mean Value Theorem] Let $f : [a, b] \to R$ be continuous on $[a, b]$ and differentiable on (a, b). There exists $c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

We construct an auxiliary function $g : [a, b] \to R$ representing the difference between $f(x)$ and the secant line passing through $(a, f(a))$ and $(b, f(b))$:

$$g(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$$

Since $f(x)$ and polynomials are continuous on $[a, b]$ and differentiable on (a, b),

$g(x)$ shares these properties. Evaluating g at the endpoints yields:

$$g(a) = f(a) - f(a) - 0 = 0$$

$$g(b) = f(b) - f(a) - (f(b) - f(a)) = 0$$

Since $g(a) = g(b) = 0$, g satisfies the hypotheses of Rolle's Theorem. Thus, there exists a point $c \in (a, b)$ such that $g'(c) = 0$. Differentiating $g(x)$ gives:

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$$

Setting $g'(c) = 0$ implies $f'(c) = \frac{f(b)-f(a)}{b-a}$, completing the proof.

[Cauchy's Mean Value Theorem] Let $f, g : [a, b] \to R$ be continuous on $[a, b]$ and differentiable on (a, b). There exists $c \in (a, b)$ such that:

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$$

Define the auxiliary function $h : [a, b] \to R$ as follows:

$$h(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x)$$

The function h is continuous on $[a, b]$ and differentiable on (a, b) due to the respective properties of f and g. Evaluating h at the endpoints:

$$h(a) = f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) = f(b)g(a) - g(b)f(a)$$

$$h(b) = f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) = g(a)f(b) - f(a)g(b)$$

Thus, $h(a) = h(b)$. By Rolle's Theorem, there exists $c \in (a, b)$ such that $h'(c) = 0$. Differentiating h yields:

$$h'(x) = (f(b) - f(a))g'(x) - (g(b) - g(a))f'(x)$$

Setting $h'(c) = 0$ directly produces $(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$.

A powerful generalization of the Mean Value Theorem arises when analyzing higher-order derivatives, leading to Taylor's Theorem, which quantifies the error in polynomial approximations of functions.

[Taylor's Theorem with Lagrange Remainder] Let $f : [a, b] \to R$ be a function such that f and its first n derivatives are continuous on $[a, b]$, and $f^{(n+1)}$ exists on (a, b). Let $x_0 \in [a, b]$. For any $x \in [a, b]$ with $x \neq x_0$, there exists a point c strictly between x and x_0 such that:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n + 1)!}(x - x_0)^{n+1}$$

Fix $x \in [a, b]$. We define a real number M such that the following equation holds:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + M(x - x_0)^{n+1}$$

Our goal is to show that $M = \frac{f^{(n+1)}(c)}{(n+1)!}$ for some c between x_0 and x. To this end, define a function $F(t)$ on the closed interval bounded by x_0 and x:

$$F(t) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x-t)^k - M(x-t)^{n+1}$$

By construction, $F(x) = f(x) - f(x) - 0 = 0$, and by our definition of M, $F(x_0) = 0$. Since F satisfies the conditions of Rolle's Theorem, there exists a point c strictly between x_0 and x such that $F'(c) = 0$.

We compute the derivative of $F(t)$ with respect to t. The summation expands as a telescoping series when differentiated using the product rule:

$$\frac{d}{dt}\left(\frac{f^{(k)}(t)}{k!}(x-t)^k\right) = \frac{f^{(k+1)}(t)}{k!}(x-t)^k - \frac{f^{(k)}(t)}{(k-1)!}(x-t)^{k-1}$$

Summing this from $k = 0$ to n, all terms cancel except the final one:

$$\frac{d}{dt}\sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x-t)^k = \frac{f^{(n+1)}(t)}{n!}(x-t)^n$$

Thus, the derivative of $F(t)$ simplifies to:

$$F'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n + M(n+1)(x-t)^n$$

Evaluating at $t = c$ and setting $F'(c) = 0$ yields:

$$0 = (x-c)^n \left(M(n+1) - \frac{f^{(n+1)}(c)}{n!}\right)$$

Since $c \neq x$, $(x-c)^n \neq 0$, which mandates:

$$M(n+1) = \frac{f^{(n+1)}(c)}{n!} M = \frac{f^{(n+1)}(c)}{(n+1)!}$$

Substituting this value of M back into the original equation for $f(x)$ completes the proof.

0.1 Riemann-Stieltjes Integrability Criteria and Measure-Zero Discontinuities

We generalize the classical Riemann integral by integrating a function f with respect to a monotonically increasing function α. This Riemann-Stieltjes framework provides the theoretical foundation for probability theory and unified functional analysis.

Let $[a, b]$ be a closed interval. A **partition** P of $[a, b]$ is a finite set of points $P = \{x_0, x_1, \ldots, x_n\}$ such that $a = x_0 < x_1 < \ldots < x_n = b$. We define $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$. For a bounded function $f : [a, b] \to R$, we define the upper and lower Darboux-Stieltjes sums with respect to P and α as:

$$U(P, f, \alpha) = \sum_{i=1}^{n} M_i \Delta\alpha_i, \quad L(P, f, \alpha) = \sum_{i=1}^{n} m_i \Delta\alpha_i$$

where $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$ and $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$.

We define the upper and lower integrals as the infimum and supremum of these sums over all possible partitions, respectively. If these two values coincide, we say f is Riemann-Stieltjes integrable with respect to α, denoted $f \in \mathcal{R}(\alpha)$.

[Cauchy Criterion for Integrability] Let f be bounded on $[a, b]$, and let α be monotonically increasing. Then $f \in \mathcal{R}(\alpha)$ if and only if for every $\epsilon > 0$, there exists a partition P such that:

$$U(P, f, \alpha) - L(P, f, \alpha) < \epsilon$$

() Assume that for every $\epsilon > 0$, there exists a partition P satisfying the inequality. By definition, the upper integral $\overline{\int} f\, d\alpha$ and lower integral $\underline{\int} f\, d\alpha$ satisfy:

$$L(P, f, \alpha) \leq \underline{\int} f\, d\alpha \leq \overline{\int} f\, d\alpha \leq U(P, f, \alpha)$$

Thus, we have:

$$0 \leq \overline{\int} f\, d\alpha - \underline{\int} f\, d\alpha \leq U(P, f, \alpha) - L(P, f, \alpha) < \epsilon$$

Since this holds for all $\epsilon > 0$, the upper and lower integrals must be equal, proving $f \in \mathcal{R}(\alpha)$.

() Assume $f \in \mathcal{R}(\alpha)$, and let $I = \int f\, d\alpha$. Let $\epsilon > 0$. By the definitions of the supremum and infimum, there exist partitions P_1 and P_2 such that:

$$U(P_1, f, \alpha) < I + \frac{\epsilon}{2} \quad and \quad L(P_2, f, \alpha) > I - \frac{\epsilon}{2}$$

Let $P = P_1 \cup P_2$ be the common refinement. Because refining a partition can only decrease upper sums and increase lower sums, we have:

$$U(P, f, \alpha) \leq U(P_1, f, \alpha) < I + \frac{\epsilon}{2}$$

$$L(P, f, \alpha) \geq L(P_2, f, \alpha) > I - \frac{\epsilon}{2}$$

Subtracting the lower sum inequality from the upper sum inequality yields:

$$U(P, f, \alpha) - L(P, f, \alpha) < \left(I + \frac{\epsilon}{2}\right) - \left(I - \frac{\epsilon}{2}\right) = \epsilon$$

To characterize precisely which functions are integrable, we must analyze their points of discontinuity. For the standard Riemann integral where $\alpha(x) = x$, Lebesgue provided an exact characterization using the concept of measure zero.

A set $E \subset R$ has **Lebesgue measure zero** if for every $\epsilon > 0$, there exists a countable collection of open intervals $\{(a_k, b_k)\}_{k=1}^{\infty}$ such that $E \subset \bigcup_{k=1}^{\infty}(a_k, b_k)$ and $\sum_{k=1}^{\infty}(b_k - a_k) < \epsilon$.

The **oscillation** of a bounded function f on an interval J is $\omega_f(J) = \sup_J f - \inf_J f$. The oscillation of f at a point x is defined as $\omega_f(x) = \lim_{\delta \to 0} \omega_f((x - \delta, x + \delta))$.

It is a standard topological result that f is continuous at x if and only if $\omega_f(x) = 0$. Furthermore, for any $\eta > 0$, the set of points $D_\eta = \{x \in [a, b] \mid \omega_f(x) \geq \eta\}$ is closed.

[Lebesgue's Criterion - Sufficiency] Let $f : [a, b] \to R$ be bounded. If the set D of discontinuities of f has Lebesgue measure zero, then f is Riemann integrable on $[a, b]$.

Let $M = \sup_{x \in [a,b]} |f(x)|$. If $M = 0$, f is identically zero and trivially integrable. Assume $M > 0$. Let $\epsilon > 0$ be given. We must construct a partition P such that $U(P, f) - L(P, f) < \epsilon$.

Let $\eta = \frac{\epsilon}{2(b-a)}$. Consider the set $D_\eta = \{x \in [a, b] \mid \omega_f(x) \geq \eta\}$. Because D_η is a closed subset of the compact interval $[a, b]$, D_η is compact. Furthermore, since $D_\eta \subset D$ and D has measure zero, D_η also has measure zero.

Therefore, there exists a countable collection of open intervals covering D_η with total length strictly less than $\frac{\epsilon}{4M}$. Because D_η is compact, we can extract a finite subcover $\mathcal{V} = \{V_1, \ldots, V_k\}$. Let $V = \bigcup_{i=1}^{k} V_i$. The sum of the lengths of the intervals in $\mathcal{V}$ is also less than $\frac{\epsilon}{4M}$.

Now consider the set $K = [a, b] \setminus V$. Since V is open, K is closed and bounded, thus compact. For every $x \in K$, we have $x \notin D_\eta$, which means $\omega_f(x) < \eta$. By the definition of oscillation at a point, there exists an open interval U_x containing x such that $\omega_f(U_x) < \eta$. The collection $\{U_x\}_{x \in K}$ forms an open cover of the compact set K. Thus, there exists a finite subcover $\mathcal{U} = \{U_{x_1}, \ldots, U_{x_m}\}$.

The endpoints of the intervals in the finite covers $\mathcal{V}$ and $\mathcal{U}$ that intersect $[a, b]$ define a partition $P = \{t_0, t_1, \ldots, t_N\}$ of $[a, b]$. We separate the subintervals $[t_{j-1}, t_j]$ into two disjoint classes:

1. Class $\mathcal{A}$ contains those subintervals that intersect D_η. By construction, these are entirely contained within V.

2. Class $\mathcal{B}$ contains all remaining subintervals. These do not intersect D_η, and are entirely contained within some interval from $\mathcal{U}$.

We now bound the difference between the upper and lower sums:

$$U(P,f) - L(P,f) = \sum_{j=1}^{N} (M_j - m_j)\Delta t_j = \sum_{j \in \mathcal{A}} (M_j - m_j)\Delta t_j + \sum_{j \in \mathcal{B}} (M_j - m_j)\Delta t_j$$

For subintervals in Class $\mathcal{A}$, the maximum difference $M_j - m_j$ is bounded trivially by $2M$. The sum of their lengths Δt_j is bounded by the total length of $\mathcal{V}$, which is less than $\frac{\epsilon}{4M}$. Thus:

$$\sum_{j \in \mathcal{A}} (M_j - m_j)\Delta t_j \le 2M \sum_{j \in \mathcal{A}} \Delta t_j < 2M \left(\frac{\epsilon}{4M} \right) = \frac{\epsilon}{2}$$

For subintervals in Class $\mathcal{B}$, each is contained in an interval where the oscillation is strictly less than η. Hence, $M_j - m_j < \eta$. The sum of their lengths is bounded by the total length of $[a,b]$, which is $b - a$. Thus:

$$\sum_{j \in \mathcal{B}} (M_j - m_j)\Delta t_j < \eta \sum_{j \in \mathcal{B}} \Delta t_j \le \eta(b-a) = \frac{\epsilon}{2(b-a)}(b-a) = \frac{\epsilon}{2}$$

Adding the contributions from both classes yields:

$$U(P,f) - L(P,f) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

By the Cauchy criterion, f is Riemann integrable on $[a,b]$.

0.1 Sequences and Series of Functions: Pointwise vs. Uniform Convergence

The transition from analyzing individual functions to sequences of functions $(f_n)_{n=1}^{\infty}$ requires a careful distinction between local limits and global limits. The mode of convergence dictates which analytic properties—such as continuity, differentiability, and integrability—are preserved in the limit function.

Let $E \subset R$, and let $(f_n)_{n=1}^{\infty}$ be a sequence of functions defined on E. We say that (f_n) **converges pointwise** to a function $f : E \to R$ if, for every $x \in E$ and every $\epsilon > 0$, there exists an integer $N \in N$ (dependent on both x and ϵ) such that for all $n \geq N$,

$$|f_n(x) - f(x)| < \epsilon$$

Pointwise convergence is often too weak to preserve analytic properties; a limit of continuous functions may be entirely discontinuous. To guarantee the preservation of continuity, we require a stronger, global form of convergence.

A sequence of functions $(f_n)_{n=1}^{\infty}$ **converges uniformly** to f on E if, for every $\epsilon > 0$, there exists an integer $N \in N$ (dependent only on ϵ) such that for all $n \geq N$ and all $x \in E$,

$$|f_n(x) - f(x)| < \epsilon$$

[Uniform Convergence and Continuity] Let (f_n) be a sequence of continuous functions on $E \subset R$. If (f_n) converges uniformly to f on E, then f is continuous on E.

Let $c \in E$ and let $\epsilon > 0$ be given. We must find a $\delta > 0$ such that for all $x \in E$ with $|x - c| < \delta$, we have $|f(x) - f(c)| < \epsilon$.

By the uniform convergence of (f_n), there exists $N \in N$ such that for all $x \in E$,

$$|f_N(x) - f(x)| < \frac{\epsilon}{3}$$

Because f_N is continuous at c, there exists a $\delta > 0$ such that for all $x \in E$ satisfying $|x - c| < \delta$,

$$|f_N(x) - f_N(c)| < \frac{\epsilon}{3}$$

We apply the triangle inequality to bound the difference $|f(x) - f(c)|$ for any $x \in E$ strictly within δ of c:

$$|f(x) - f(c)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)|$$

Substituting our bounds yields:

$$|f(x) - f(c)| < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

Thus, f is continuous at c. Since c was arbitrary, f is continuous on the entirety of E.

When analyzing infinite series of functions, $\sum_{n=1}^{\infty} f_n(x)$, uniform convergence is most efficiently established via a bounding criterion over the summands.

[Weierstrass M-Test] Let (f_n) be a sequence of functions defined on E, and suppose there exists a sequence of real numbers (M_n) such that $|f_n(x)| \leq M_n$ for all $x \in E$ and all $n \in N$. If the infinite series $\sum_{n=1}^{\infty} M_n$ converges, then the series $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on E.

Let $S_k(x) = \sum_{n=1}^{k} f_n(x)$ denote the sequence of partial sums. We will show that (S_k) satisfies the uniform Cauchy criterion. Let $\epsilon > 0$. Because $\sum_{n=1}^{\infty} M_n$ converges, the sequence of its partial sums is Cauchy. Thus, there exists an $N \in N$ such that for all $m > k \geq N$,

$$\sum_{n=k+1}^{m} M_n < \epsilon$$

For any $x \in E$ and for $m > k \geq N$, the triangle inequality implies:

$$|S_m(x) - S_k(x)| = \left| \sum_{n=k+1}^{m} f_n(x) \right| \leq \sum_{n=k+1}^{m} |f_n(x)| \leq \sum_{n=k+1}^{m} M_n < \epsilon$$

Since N depends strictly on ϵ and is entirely independent of x, the sequence of partial sums (S_k) is uniformly Cauchy. Therefore, the series converges uniformly to a limit function on E.

[Integration of Uniformly Convergent Sequences] Let α be monotonically increasing on $[a, b]$. Suppose $f_n \in \mathcal{R}(\alpha)$ on $[a, b]$ for $n = 1, 2, \ldots$, and suppose $f_n \to f$ uniformly on $[a, b]$. Then $f \in \mathcal{R}(\alpha)$ on $[a, b]$, and:

$$\int_a^b f \, d\alpha = \lim_{n \to \infty} \int_a^b f_n \, d\alpha$$

Let $\epsilon > 0$. Choose a constant $\epsilon_0 > 0$ such that $\epsilon_0(\alpha(b) - \alpha(a)) < \frac{\epsilon}{3}$. By the definition of uniform convergence, there exists $N \in N$ such that for all $n \geq N$ and $x \in [a, b]$,

$$f_n(x) - \epsilon_0 < f(x) < f_n(x) + \epsilon_0$$

This strict bounding implies that for the upper and lower Darboux-Stieltjes integrals:

$$\overline{\int} f \, d\alpha \leq \overline{\int} (f_n + \epsilon_0) \, d\alpha = \int_a^b f_n \, d\alpha + \epsilon_0(\alpha(b) - \alpha(a))$$

$$\underline{\int} f \, d\alpha \geq \underline{\int} (f_n - \epsilon_0) \, d\alpha = \int_a^b f_n \, d\alpha - \epsilon_0(\alpha(b) - \alpha(a))$$

Therefore, the difference between the upper and lower integrals of f is strictly bounded by:

$$0 \leq \overline{\int} f \, d\alpha - \underline{\int} f \, d\alpha \leq 2\epsilon_0(\alpha(b) - \alpha(a)) < \frac{2\epsilon}{3} < \epsilon$$

Since ϵ was arbitrary, the upper and lower integrals coincide: $\overline{\int} f\,d\alpha = \underline{\int} f\,d\alpha$, proving $f \in \mathcal{R}(\alpha)$. Furthermore, evaluating the difference of the integrals for $n \geq N$ gives:

$$\left|\int_a^b f\,d\alpha - \int_a^b f_n\,d\alpha\right| = \left|\int_a^b (f - f_n)\,d\alpha\right| \leq \int_a^b |f - f_n|\,d\alpha \leq \epsilon_0(\alpha(b) - \alpha(a)) < \epsilon$$

This proves that the sequence of integrals converges to the integral of the limit function.

1 Linear Algebra & Spectral Theory

1.1 Vector Spaces, Linear Maps, and the Rank-Nullity Theorem

We shift our theoretical focus to the algebraic structures underlying vector spaces, providing the formal framework necessary for spectral theory. Let F denote an arbitrary field, which in the context of this portfolio will typically be R or C.

A **vector space** V over a field F is a set equipped with two operations: vector addition $+ : V \times V \to V$ and scalar multiplication $\cdot : F \times V \to V$. These operations must satisfy commutativity and associativity of addition, the existence of an additive identity 0_V and additive inverses, the existence of a multiplicative identity $1 \in F$, and distributivity of scalar multiplication over vector and scalar addition.

Let V and W be vector spaces over the same field F. A function $T : V \to W$ is a **linear map** if for all $u, v \in V$ and all $c \in F$:

$$T(u + v) = T(u) + T(v) \quad and \quad T(cu) = cT(u)$$

Associated with any linear map $T : V \to W$ are two fundamental subsets that dictate its injectivity and surjectivity.

The **kernel** (or null space) of T is $\ker(T) = \{v \in V \mid T(v) = 0_W\}$. The **image** (or range) of T is $im(T) = \{T(v) \in W \mid v \in V\}$.

For any linear map $T : V \to W$, $\ker(T)$ is a subspace of V, and $im(T)$ is a subspace of W.

We first prove $\ker(T)$ is a subspace. By linearity, $T(0_V) = T(0_V + 0_V) = T(0_V) + T(0_V)$, which forces $T(0_V) = 0_W$. Thus, $0_V \in \ker(T)$, making it non-empty. Let $u, v \in \ker(T)$ and $c \in F$. By linearity, $T(u + v) = T(u) + T(v) = 0_W + 0_W = 0_W$, and $T(cu) = cT(u) = c(0_W) = 0_W$. Because it is closed under vector addition and scalar multiplication, $\ker(T)$ is a subspace of V.

Next, we prove $im(T)$ is a subspace. Since $T(0_V) = 0_W$, $0_W \in im(T)$. Let $w_1, w_2 \in im(T)$ and $c \in F$. By definition, there exist $v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. Because V is a vector space, $v_1 + v_2 \in V$ and $cv_1 \in V$. Applying T yields $T(v_1 + v_2) = T(v_1) + T(v_2) = w_1 + w_2$, and $T(cv_1) = cT(v_1) = cw_1$. Thus, $im(T)$ is closed under addition and scalar multiplication, constituting a subspace of W.

[Rank-Nullity Theorem] Let V and W be vector spaces over F, and let V be finite-dimensional. If $T : V \to W$ is a linear map, then:

$$\dim(\ker(T)) + \dim(im(T)) = \dim(V)$$

Let $\dim(V) = n$ and let $k = \dim(\ker(T))$. Since $\ker(T)$ is a subspace of the finite-dimensional vector space V, it is finite-dimensional, and $0 \le k \le n$. Let $\{u_1, u_2, \ldots, u_k\}$ be a basis for $\ker(T)$.

By the Basis Extension Theorem, any linearly independent set in a finite-dimensional vector space can be extended to a basis for the entire space. Therefore, we can append $r = n - k$ vectors $\{v_1, v_2, \ldots, v_r\}$ such that the combined set:

$$\mathcal{B} = \{u_1, \dots, u_k, v_1, \dots, v_r\}$$

forms a basis for V. To prove the theorem, it suffices to show that the set $\mathcal{C} = \{T(v_1), T(v_2), \dots, T(v_r)\}$ forms a basis for $im(T)$, which requires proving that $\mathcal{C}$ both spans $im(T)$ and is linearly independent.

First, we establish that $\mathcal{C}$ spans $im(T)$. Let $w \in im(T)$. By definition, there exists $x \in V$ such that $T(x) = w$. Because $\mathcal{B}$ is a basis for V, we express x uniquely as a linear combination:

$$x = \sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{r} b_j v_j$$

for scalars $a_i, b_j \in F$. Applying the linear map T yields:

$$w = T(x) = T\left(\sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{r} b_j v_j\right) = \sum_{i=1}^{k} a_i T(u_i) + \sum_{j=1}^{r} b_j T(v_j)$$

Because $u_i \in \ker(T)$ for all $1 \leq i \leq k$, $T(u_i) = 0_W$. The sum truncates to:

$$w = \sum_{j=1}^{r} b_j T(v_j)$$

This demonstrates that any $w \in im(T)$ can be written as a linear combination of vectors in $\mathcal{C}$, meaning $span(\mathcal{C}) = im(T)$.

Second, we prove that $\mathcal{C}$ is linearly independent. Suppose there exist scalars $c_1, \dots, c_r \in F$ such that:

$$\sum_{j=1}^{r} c_j T(v_j) = 0_W$$

By the linearity of T, we consolidate the left-hand side:

$$T\left(\sum_{j=1}^{r} c_j v_j\right) = 0_W$$

This implies that the vector $y = \sum_{j=1}^{r} c_j v_j$ lies in $\ker(T)$. Because $\{u_1, \dots, u_k\}$ is a basis for $\ker(T)$, y must be expressible as a linear combination of the u_i basis vectors:

$$\sum_{j=1}^{r} c_j v_j = \sum_{i=1}^{k} d_i u_i \sum_{i=1}^{k} d_i u_i - \sum_{j=1}^{r} c_j v_j = 0_V$$

The set $\mathcal{B} = \{u_1, \dots, u_k, v_1, \dots, v_r\}$ is a basis for V, meaning it is strictly linearly independent. The only solution to this linear combination evaluating to the zero vector is the trivial solution. Thus, $c_j = 0$ for all $1 \leq j \leq r$.

Because the only solution to $\sum_{j=1}^{r} c_j T(v_j) = 0_W$ is the trivial one, the set $\mathcal{C}$ is linearly independent. Since $\mathcal{C}$ is a linearly independent spanning set of size r, it is a basis for $im(T)$. Therefore, $\dim(im(T)) = r$.

Summing the dimensions concludes the proof:

$$\dim(\ker(T)) + \dim(im(T)) = k + r = k + (n - k) = n = \dim(V)$$

0.1 Eigenvalues, Eigenvectors, and the Spectral Theorem for Symmetric/Hermitian Matrices

To understand the intrinsic geometric behavior of a linear operator $T : V \to V$, we isolate the specific directions in V that are strictly scaled by T, invariant in their span. We assume V is a finite-dimensional inner product space over the field F (where F is either R or C), equipped with an inner product $\langle \cdot, \cdot \rangle$.

Let $T : V \to V$ be a linear operator. A scalar $\lambda \in F$ is an **eigenvalue** of T if there exists a non-zero vector $v \in V$ such that $T(v) = \lambda v$. The vector v is called an **eigenvector** corresponding to λ.

The **adjoint** of a linear operator $T \in \mathcal{L}(V)$, denoted T^*, is the unique linear operator on V such that $\langle T(u), v \rangle = \langle u, T^*(v) \rangle$ for all $u, v \in V$. An operator T is **self-adjoint** (or Hermitian) if $T = T^*$. In the context of real matrix representations, this is equivalent to a symmetric matrix ($A = A^T$).

The structure of self-adjoint operators is remarkably rigid. We first prove two foundational lemmas regarding their spectra.

[Real Spectrum of Self-Adjoint Operators] If $T : V \to V$ is a self-adjoint operator, then all eigenvalues of T are strictly real.

Let $\lambda \in C$ be an eigenvalue of T, and let $v \in V$ be a corresponding non-zero eigenvector. We evaluate the inner product $\langle T(v), v \rangle$ in two distinct ways. On one hand, applying the eigenvalue definition directly yields:

$$\langle T(v), v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle$$

On the other hand, utilizing the self-adjoint property $T = T^*$:

$$\langle T(v), v \rangle = \langle v, T^*(v) \rangle = \langle v, T(v) \rangle = \langle v, \lambda v \rangle = \overline{\lambda} \langle v, v \rangle$$

Equating the two results gives $\lambda \langle v, v \rangle = \overline{\lambda} \langle v, v \rangle$. Since v is a non-zero eigenvector, $\langle v, v \rangle = \|v\|^2 \neq 0$. Dividing by $\|v\|^2$ yields $\lambda = \overline{\lambda}$, proving that $\lambda \in R$.

[Orthogonality of Eigenspaces] Let $T : V \to V$ be a self-adjoint operator. If λ_1 and λ_2 are distinct eigenvalues of T, then their corresponding eigenvectors v_1 and v_2 are orthogonal.

Suppose $T(v_1) = \lambda_1 v_1$ and $T(v_2) = \lambda_2 v_2$, with $\lambda_1 \neq \lambda_2$. We evaluate the inner product $\langle T(v_1), v_2 \rangle$:

$$\langle T(v_1), v_2 \rangle = \langle \lambda_1 v_1, v_2 \rangle = \lambda_1 \langle v_1, v_2 \rangle$$

Alternatively, utilizing the adjoint property and the fact that eigenvalues of T are real (hence $\overline{\lambda_2} = \lambda_2$):

$$\langle T(v_1), v_2 \rangle = \langle v_1, T^*(v_2) \rangle = \langle v_1, T(v_2) \rangle = \langle v_1, \lambda_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle$$

Equating these expansions gives $\lambda_1 \langle v_1, v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle$, which rearranges to:

$$(\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle = 0$$

Because $\lambda_1 \neq \lambda_2$, we must have $\langle v_1, v_2 \rangle = 0$, proving that v_1 and v_2 are orthogonal.

We are now equipped to prove the central result of finite-dimensional spectral theory, which demonstrates that self-adjoint operators can be completely decoupled into independent orthogonal 1-dimensional projections.

[Spectral Theorem for Finite-Dimensional Spaces] Let V be a finite-dimensional inner product space over F, and let $T \in \mathcal{L}(V)$ be a self-adjoint operator. Then V admits an orthonormal basis consisting entirely of eigenvectors of T.

We proceed by mathematical induction on $n = \dim(V)$.

Base Case: If $n = 1$, any non-zero vector $v \in V$ is an eigenvector of T. Normalizing v by setting $e_1 = \frac{v}{\|v\|}$ provides the required orthonormal basis. The theorem trivially holds.

Inductive Step: Assume the theorem is true for all inner product spaces of dimension $n-1$. Let $\dim(V) = n$. By the Fundamental Theorem of Algebra, the characteristic polynomial of T possesses at least one root over C. Because T is self-adjoint, Lemma 3.2.1 guarantees this root is real, ensuring T has at least one real eigenvalue λ_1 and a corresponding non-zero eigenvector v_1. We normalize this vector to obtain a unit eigenvector $e_1 = \frac{v_1}{\|v_1\|}$.

Define $W = span(e_1)^\perp = \{u \in V \mid \langle u, e_1 \rangle = 0\}$. Since $\dim(V) = n$ and $\dim(span(e_1)) = 1$, standard properties of orthogonal complements dictate that $\dim(W) = n - 1$.

We claim that W is invariant under T (i.e., $T(W) \subset W$). Let $w \in W$. We must show that $T(w) \in W$, which requires $\langle T(w), e_1 \rangle = 0$. Using the self-adjointness of T and the fact that $T(e_1) = \lambda_1 e_1$:

$$\langle T(w), e_1 \rangle = \langle w, T^*(e_1) \rangle = \langle w, T(e_1) \rangle = \langle w, \lambda_1 e_1 \rangle = \lambda_1 \langle w, e_1 \rangle$$

Because $w \in W = span(e_1)^\perp$, $\langle w, e_1 \rangle = 0$. Thus, $\langle T(w), e_1 \rangle = 0$, proving $T(w) \in W$. Therefore, the restriction of T to W, denoted $T|_W : W \to W$, is a well-defined linear operator.

Furthermore, $T|_W$ inherits self-adjointness from T. For any $x, y \in W$, $\langle T|_W(x), y \rangle = \langle T(x), y \rangle = \langle x, T(y) \rangle = \langle x, T|_W(y) \rangle$.

Because $T|_W$ is a self-adjoint operator on the $(n-1)$-dimensional inner product space W, the inductive hypothesis applies. There exists an orthonormal basis $\{e_2, e_3, \ldots, e_n\}$ for W consisting of eigenvectors of $T|_W$ (and consequently, eigenvectors of T).

We construct the set $\mathcal{B} = \{e_1, e_2, \ldots, e_n\}$. Since e_1 is orthogonal to every vector in W, and $\{e_2, \ldots, e_n\}$ is an orthonormal set in W, $\mathcal{B}$ forms an orthonormal set of n vectors in the n-dimensional space V. Thus, $\mathcal{B}$ is an orthonormal basis for V composed entirely of eigenvectors of T, completing the induction.

0.1 Singular Value Decomposition (SVD) and Geometric Projections

While the Spectral Theorem strictly applies to self-adjoint operators on a single vector space, the Singular Value Decomposition (SVD) generalizes this geometric decoupling to arbitrary linear maps between vector spaces of potentially different dimensions.

Let V and W be finite-dimensional inner product spaces over F (where F is R or C). Consider a linear map $T : V \to W$. Its adjoint $T^* : W \to V$ is uniquely defined by the relation $\langle T(v), w\rangle_W = \langle v, T^*(w)\rangle_V$ for all $v \in V$ and $w \in W$.

The composite operator $T^*T : V \to V$ is self-adjoint and positive semi-definite. Consequently, its eigenvalues are real and non-negative.

To prove self-adjointness, we take the adjoint of the composition. By the properties of adjoints, $(T^*T)^* = T^*(T^*)^* = T^*T$. Thus, T^*T is self-adjoint.

To prove it is positive semi-definite, let $v \in V$. We evaluate the inner product:

$$\langle T^*T(v), v\rangle_V = \langle T(v), T(v)\rangle_W = \|T(v)\|_W^2$$

By the axioms of norms, $\|T(v)\|_W^2 \geq 0$. If λ is an eigenvalue of T^*T with corresponding unit eigenvector v, then $T^*T(v) = \lambda v$, which implies:

$$\lambda = \lambda\langle v, v\rangle_V = \langle \lambda v, v\rangle_V = \langle T^*T(v), v\rangle_V = \|T(v)\|_W^2 \geq 0$$

Thus, all eigenvalues of T^*T are real and non-negative.

Let $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n \geq 0$ be the eigenvalues of T^*T, counted with multiplicity. The **singular values** of T, denoted σ_i, are the non-negative square roots of these eigenvalues: $\sigma_i = \sqrt{\lambda_i}$.

[Singular Value Decomposition] Let $T : V \to W$ be a linear map with rank r. There exist orthonormal bases $\{v_1, \ldots, v_n\}$ of V and $\{u_1, \ldots, u_m\}$ of W, and strictly positive singular values $\sigma_1 \geq \ldots \geq \sigma_r > 0$, such that for all $v \in V$:

$$T(v) = \sum_{i=1}^{r} \sigma_i \langle v, v_i\rangle_V u_i$$

Because T^*T is a self-adjoint operator on the finite-dimensional space V, the Spectral Theorem (Theorem 3.2.3) guarantees the existence of an orthonormal basis $\{v_1, \ldots, v_n\}$ of V consisting of eigenvectors of T^*T. Order these eigenvectors such that their corresponding eigenvalues satisfy $\lambda_1 \geq \ldots \geq \lambda_r > 0$ and $\lambda_{r+1} = \ldots = \lambda_n = 0$. The singular values are $\sigma_i = \sqrt{\lambda_i}$.

For $1 \leq i \leq r$, define vectors $u_i \in W$ by:

$$u_i = \frac{1}{\sigma_i} T(v_i)$$

We first demonstrate that the set $\{u_1, \ldots, u_r\}$ is orthonormal in W. For any $1 \leq i, j \leq r$, we compute the inner product:

$$\langle u_i, u_j\rangle_W = \left\langle \frac{1}{\sigma_i} T(v_i), \frac{1}{\sigma_j} T(v_j) \right\rangle_W = \frac{1}{\sigma_i \sigma_j} \langle T^*T(v_i), v_j\rangle_V$$

Since v_i is an eigenvector of T^*T with eigenvalue $\lambda_i = \sigma_i^2$, this becomes:

$$\langle u_i, u_j \rangle_W = \frac{\sigma_i^2}{\sigma_i \sigma_j} \langle v_i, v_j \rangle_V$$

Because $\{v_1, \dots, v_n\}$ is an orthonormal basis, $\langle v_i, v_j \rangle_V = \delta_{ij}$ (the Kronecker delta). Thus, $\langle u_i, u_j \rangle_W = \delta_{ij}$, proving $\{u_1, \dots, u_r\}$ is an orthonormal set. We can extend this set to a full orthonormal basis $\{u_1, \dots, u_m\}$ for W using the Gram-Schmidt process.

To verify the decomposition, let $v \in V$. We express v in terms of the orthonormal basis of V:

$$v = \sum_{i=1}^{n} \langle v, v_i \rangle_V v_i$$

Applying the linear map T yields:

$$T(v) = \sum_{i=1}^{n} \langle v, v_i \rangle_V T(v_i)$$

For $i > r$, $\lambda_i = 0$, which implies $\|T(v_i)\|_W^2 = \langle T^*T(v_i), v_i \rangle_V = 0$, forcing $T(v_i) = 0_W$. Therefore, the sum truncates at r. Substituting $T(v_i) = \sigma_i u_i$ for $1 \leq i \leq r$ gives the desired representation:

$$T(v) = \sum_{i=1}^{r} \sigma_i \langle v, v_i \rangle_V u_i$$

1 Metric Spaces & General Topology

1.1 Metric Spaces, Topological Properties, and Completeness

We now abstract the topological concepts developed for R in Chapter 1 to arbitrary sets equipped with a notion of distance. This abstraction provides the theoretical foundation for functional analysis, where the points in the space are often themselves functions.

A **metric space** is a pair (X, d), where X is a set and $d : X \times X \to R$ is a function (the metric) satisfying the following properties for all $x, y, z \in X$:

1. **Positivity:** $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
2. **Symmetry:** $d(x, y) = d(y, x)$.
3. **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

The topology of a metric space is generated by open balls. For $x \in X$ and $r > 0$, the open ball of radius r centered at x is defined as $B_r(x) = \{y \in X \mid d(x, y) < r\}$. A subset $U \subset X$ is **open** if for every $x \in U$, there exists an $r > 0$ such that $B_r(x) \subset U$. A subset $F \subset X$ is **closed** if its complement $X \setminus F$ is open.

A sequence $(x_n)_{n=1}^{\infty}$ in a metric space (X, d) is a **Cauchy sequence** if for every $\epsilon > 0$, there exists an $N \in N$ such that for all $m, n \geq N$, $d(x_m, x_n) < \epsilon$. The metric space (X, d) is **complete** if every Cauchy sequence in X converges to a limit in X.

A particularly important class of complete metric spaces arises from vector spaces equipped with a norm.

A **Banach space** is a vector space V over R or C equipped with a norm $\|\cdot\|$ such that the metric space induced by the metric $d(u, v) = \|u - v\|$ is complete.

Completeness is a powerful property because it guarantees the existence of a limit based solely on the internal behavior of the sequence, without prior knowledge of the limit point. This principle is exploited in one of the most fundamental results in non-linear analysis: the Banach Fixed-Point Theorem.

Let (X, d) be a metric space. A map $T : X \to X$ is a **contraction mapping** if there exists a constant $c \in [0, 1)$ such that for all $x, y \in X$:

$$d(T(x), T(y)) \leq c\, d(x, y)$$

[Banach Fixed-Point Theorem] Let (X, d) be a non-empty complete metric space, and let $T : X \to X$ be a contraction mapping with contraction constant $c \in [0, 1)$. Then T admits a unique fixed point $x^* \in X$ (i.e., $T(x^*) = x^*$).

We first prove uniqueness. Suppose $x, y \in X$ are both fixed points of T, such that $T(x) = x$ and $T(y) = y$. Applying the contraction property yields:

$$d(x, y) = d(T(x), T(y)) \leq c\, d(x, y)$$

This implies $(1-c)d(x,y) \leq 0$. Since $c < 1$, we have $1 - c > 0$. Because distance is non-negative ($d(x,y) \geq 0$), the only way this inequality can hold is if $d(x,y) = 0$. By the positivity axiom of a metric space, $x = y$. Thus, if a fixed point exists, it is strictly unique.

To prove existence, we construct a sequence via Picard iteration. Choose an arbitrary initial point $x_0 \in X$. Define the sequence $(x_n)_{n=0}^{\infty}$ recursively by $x_n = T(x_{n-1})$ for all $n \geq 1$. We must show that (x_n) is a Cauchy sequence.

Consider the distance between consecutive terms. By repeated application of the contraction property:

$$d(x_{n+1}, x_n) = d(T(x_n), T(x_{n-1})) \leq c\, d(x_n, x_{n-1}) \leq \ldots \leq c^n d(x_1, x_0)$$

Let $m > n \geq 0$. We apply the triangle inequality repeatedly to bound the distance between x_m and x_n:

$$d(x_m, x_n) \leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \ldots + d(x_{n+1}, x_n)$$

Substituting the bound for consecutive terms, we obtain:

$$d(x_m, x_n) \leq (c^{m-1} + c^{m-2} + \ldots + c^n) d(x_1, x_0) = c^n \sum_{k=0}^{m-n-1} c^k\, d(x_1, x_0)$$

Because $c \in [0,1)$, the infinite geometric series $\sum_{k=0}^{\infty} c^k$ converges to $\frac{1}{1-c}$. Therefore, we can strictly bound the finite sum:

$$d(x_m, x_n) \leq c^n \frac{1}{1-c} d(x_1, x_0)$$

Let $\epsilon > 0$. Since $c < 1$, $\lim_{n\to\infty} c^n = 0$. Thus, there exists an $N \in N$ such that for all $n \geq N$:

$$c^n < \frac{\epsilon(1-c)}{d(x_1, x_0)}$$

(If $d(x_1, x_0) = 0$, the sequence is constant and trivially Cauchy). For all $m > n \geq N$, this forces $d(x_m, x_n) < \epsilon$. Thus, (x_n) is a Cauchy sequence.

Because X is a complete metric space, the Cauchy sequence (x_n) converges to a limit $x^* \in X$. We now verify that x^* is a fixed point. A contraction mapping is globally uniformly continuous (letting $\delta = \epsilon$ works for any $c < 1$). Because T is continuous, the limit commutes with the function application:

$$T(x^*) = T\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} T(x_n) = \lim_{n\to\infty} x_{n+1} = x^*$$

Hence, x^* is a fixed point of T, completing the proof.

0.1 Compactness Variants: Total Boundedness, Sequential Compactness, and the Ascoli-ArzelTheorem

In the standard topology of R^n, the Heine-Borel theorem establishes that a set is compact if and only if it is closed and bounded. However, in arbitrary metric spaces, mere boundedness is insufficient to guarantee compactness. We must introduce a stricter geometric constraint.

A metric space (X, d) is **totally bounded** if for every $\epsilon > 0$, there exists a finite set of points $\{x_1, x_2, \ldots, x_n\} \subset X$ (an ϵ-net) such that $X = \bigcup_{i=1}^{n} B_\epsilon(x_i)$.

A metric space (X, d) is **sequentially compact** if every sequence $(x_n)_{n=1}^\infty$ in X possesses a convergent subsequence whose limit lies in X.

If a metric space (X, d) is sequentially compact, then it is totally bounded.

We proceed by contradiction. Assume (X, d) is sequentially compact but not totally bounded. Then there exists some $\epsilon_0 > 0$ for which no finite ϵ_0-net can cover X.

We construct a sequence (x_n) recursively. Choose $x_1 \in X$ arbitrarily. Because the single ball $B_{\epsilon_0}(x_1)$ cannot cover X, there exists $x_2 \in X \setminus B_{\epsilon_0}(x_1)$. This implies $d(x_1, x_2) \geq \epsilon_0$. Inductively, suppose we have chosen points $\{x_1, \ldots, x_k\}$ such that $d(x_i, x_j) \geq \epsilon_0$ for all $i \neq j$. Because no finite union of ϵ_0-balls covers X, the set $X \setminus \bigcup_{i=1}^{k} B_{\epsilon_0}(x_i)$ is non-empty. We select x_{k+1} from this complement, ensuring $d(x_{k+1}, x_i) \geq \epsilon_0$ for all $1 \leq i \leq k$.

This process yields an infinite sequence $(x_n)_{n=1}^\infty$ satisfying $d(x_n, x_m) \geq \epsilon_0$ for all $n \neq m$. Consequently, no subsequence can be a Cauchy sequence, as the distance between any two distinct terms is bounded below by ϵ_0. Since every convergent sequence must be Cauchy, (x_n) admits no convergent subsequence. This directly contradicts the premise that X is sequentially compact. Therefore, X must be totally bounded.

A central application of these metric space topologies lies in the analysis of function spaces. Let K be a compact metric space, and let $C(K)$ denote the Banach space of continuous real-valued functions on K, equipped with the supremum norm $\|f\|_\infty = \sup_{x \in K} |f(x)|$. To determine when a sequence of functions in $C(K)$ admits a uniformly convergent subsequence, we require the notions of uniform boundedness and equicontinuity.

A family of functions $\mathcal{F} \subset C(K)$ is **uniformly bounded** if there exists $M > 0$ such that $|f(x)| \leq M$ for all $f \in \mathcal{F}$ and all $x \in K$. The family $\mathcal{F}$ is **equicontinuous** if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in K$ with $d(x, y) < \delta$, we have $|f(x) - f(y)| < \epsilon$ for all $f \in \mathcal{F}$.

[Ascoli-ArzelTheorem - Sufficiency] Let K be a compact metric space. If a sequence of functions $(f_n)_{n=1}^\infty$ in $C(K)$ is uniformly bounded and equicontinuous, then it contains a uniformly convergent subsequence.

Because K is compact, it is totally bounded and thus separable; it possesses a countable dense subset $D = \{x_1, x_2, \ldots\}$.

We employ Cantor's diagonal argument. Because the sequence is uniformly bounded, the sequence of real numbers $(f_n(x_1))$ is bounded. By the Bolzano-Weierstrass theorem, it has a convergent subsequence, which we index as $(f_{1,k})_{k=1}^\infty$. Next, the sequence $(f_{1,k}(x_2))$ is a bounded real sequence, yielding a convergent subsequence $(f_{2,k})_{k=1}^{\infty}$. Proceeding inductively, we construct nested subsequences $(f_{m,k})_{k=1}^{\infty}$ such that the m-th subsequence converges at the point x_m.

We extract the diagonal sequence $g_k = f_{k,k}$. For any fixed m, $(g_k)_{k\geq m}$ is a subsequence of $(f_{m,k})_{k=1}^{\infty}$. Therefore, the sequence (g_k) converges pointwise at every $x \in D$.

We now prove that (g_k) is uniformly Cauchy on K. Let $\epsilon > 0$. By the equicontinuity of the original sequence, there exists $\delta > 0$ such that $d(x,y) < \delta |g_k(x) - g_k(y)| < \frac{\epsilon}{3}$ for all $k \in N$ and $x, y \in K$. Because K is compact, it is totally bounded. The open balls $\{B_\delta(x_i)\}_{x_i \in D}$ cover K, so we can extract a finite subcover corresponding to points $\{p_1, \ldots, p_r\} \subset D$.

Since (g_k) converges at each p_i, it is a Cauchy sequence at each of these finitely many points. Thus, there exists an $N \in N$ such that for all $m, k \geq N$ and all $1 \leq i \leq r$:

$$|g_k(p_i) - g_m(p_i)| < \frac{\epsilon}{3}$$

Let $x \in K$ be arbitrary. The point x must lie in $B_\delta(p_i)$ for some $1 \leq i \leq r$. For $m, k \geq N$, we apply the triangle inequality:

$$|g_k(x) - g_m(x)| \leq |g_k(x) - g_k(p_i)| + |g_k(p_i) - g_m(p_i)| + |g_m(p_i) - g_m(x)|$$

Since $d(x, p_i) < \delta$, equicontinuity strictly bounds the first and third terms by $\frac{\epsilon}{3}$. The middle term is bounded by $\frac{\epsilon}{3}$ due to the pointwise Cauchy convergence at p_i. Therefore:

$$|g_k(x) - g_m(x)| < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

Since this bound is independent of x, (g_k) is uniformly Cauchy. Because $C(K)$ is complete under the supremum norm, (g_k) converges uniformly on K.

0.1 Connectedness, Path-Connectedness, and the Preservation of Topological Properties

Unlike compactness, which relates to bounding and covering, connectedness describes the structural indivisibility of a space. We define this property negatively, via the inability to separate the space into disjoint open components.

A metric space (X, d) is **disconnected** if there exist two non-empty, disjoint open sets $U, V \subset X$ such that $X = U \cup V$. The space (X, d) is **connected** if it is not disconnected. A subset $E \subset X$ is connected if it is connected as a metric subspace under the inherited metric.

A more intuitive, kinesthetic notion of connectedness relies on the existence of continuous paths between any two points.

A metric space (X, d) is **path-connected** if for every pair of points $x, y \in X$, there exists a continuous function $\gamma : [0, 1] \to X$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Every path-connected metric space (X, d) is connected.

We proceed by contradiction. Assume (X, d) is path-connected but disconnected. By definition, there exist non-empty, disjoint open sets U and V such that $X = U \cup V$. Because U and V are non-empty, we can choose points $x \in U$ and $y \in V$.

Since X is path-connected, there exists a continuous path $\gamma : [0, 1] \to X$ such that $\gamma(0) = x$ and $\gamma(1) = y$. Consider the preimages of U and V under γ:

$$A = \gamma^{-1}(U) = \{t \in [0, 1] \mid \gamma(t) \in U\}$$

$$B = \gamma^{-1}(V) = \{t \in [0, 1] \mid \gamma(t) \in V\}$$

Because γ is a continuous mapping, the preimages of open sets are open in the relative topology of $[0, 1]$. Thus, A and B are open subsets of $[0, 1]$. Furthermore, $0 \in A$ (since $\gamma(0) = x \in U$) and $1 \in B$ (since $\gamma(1) = y \in V$), making both sets non-empty. Since $U \cap V = \emptyset$, it strictly follows that $A \cap B = \emptyset$. Finally, because $X = U \cup V$, every $t \in [0, 1]$ maps into either U or V, meaning $A \cup B = [0, 1]$.

This implies that $[0, 1]$ is the union of two non-empty, disjoint open sets A and B, which means $[0, 1]$ is disconnected. However, the interval $[0, 1] \subset R$ is fundamentally connected (a direct consequence of the Completeness Axiom established in Chapter 1). This contradiction forces us to conclude that X must be connected.

The power of topological invariants lies in their preservation under continuous mappings. Connectedness is perfectly preserved, which immediately yields the generalized Intermediate Value Theorem.

[Preservation of Connectedness] Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a continuous function. If X is connected, then the image $f(X)$ is connected in Y.

Suppose, for the sake of contradiction, that $f(X)$ is disconnected. Then there exist open sets U_Y and V_Y in Y such that $U = U_Y \cap f(X)$ and $V = V_Y \cap f(X)$ are non-empty, disjoint open sets in the subspace topology of $f(X)$, and $f(X) = U \cup V$.

Consider the preimages in X:

$$A = f^{-1}(U) \quad and \quad B = f^{-1}(V)$$

Because f is continuous, A and B are open in X. Since U and V are non-empty and contained in the image of f, both A and B are non-empty. Because $U \cap V = \emptyset$, a single point in X cannot map to both U and V, rendering $A \cap B = \emptyset$. Lastly, because $f(X) = U \cup V$, every point in X maps to either U or V, so $X = A \cup B$. This constitutes a disconnection of X, contradicting the hypothesis that X is connected. Thus, $f(X)$ must be connected.

[Generalized Intermediate Value Theorem] Let (X, d) be a connected metric space, and let $f : X \to R$ be continuous. If $a, b \in X$ such that $f(a) < f(b)$, then for every real number c satisfying $f(a) < c < f(b)$, there exists a point $x \in X$ such that $f(x) = c$.

By Theorem 4.3.2, the image $f(X) \subset R$ is connected. The only connected subsets of the real line are intervals (rays, segments, or R itself). Since $f(a) \in f(X)$ and $f(b) \in f(X)$, the entire segment $[f(a), f(b)]$ must be contained within the interval $f(X)$. Because $c \in [f(a), f(b)]$, it follows that $c \in f(X)$. By the definition of the image, there exists at least one $x \in X$ such that $f(x) = c$.

1 Complex Analysis & Contour Integration

1.1 Complex Differentiability, the Cauchy-Riemann Equations, and Analytic Structure

We now extend the calculus of real variables to the complex plane, C. Identifying C topologically with R^2 under the standard Euclidean metric, we analyze functions of a complex variable $z = x + iy$. The requirement of complex differentiability is vastly more rigid than its real counterpart, imposing profound structural symmetry on the component functions.

Let $\Omega \subset C$ be an open set, and let $f : \Omega \to C$ be a complex-valued function. The function f is **complex differentiable** at a point $z_0 \in \Omega$ if the following limit exists:

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

where $h \in C$ approaches 0 along any arbitrary path in the complex plane. If f is complex differentiable at every point in an open set Ω, it is said to be **holomorphic** (or analytic) on Ω.

To understand the geometric severity of this definition, we decompose f into its real and imaginary parts: $f(z) = f(x + iy) = u(x, y) + iv(x, y)$, where u and v are real-valued functions of two real variables.

[The Cauchy-Riemann Equations - Necessity] If $f(z) = u(x, y) + iv(x, y)$ is complex differentiable at $z_0 = x_0 + iy_0$, then the first-order partial derivatives of u and v exist at (x_0, y_0) and satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad and \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Assume $f'(z_0)$ exists. We evaluate the limit along two orthogonal trajectories.

First, let h approach 0 along the real axis. We write $h = \Delta x + i0$, where $\Delta x \in R$ and $\Delta x \to 0$. The difference quotient becomes:

$$f'(z_0) = \lim_{\Delta x \to 0} \frac{u(x_0 + \Delta x, y_0) + iv(x_0 + \Delta x, y_0) - (u(x_0, y_0) + iv(x_0, y_0))}{\Delta x}$$

Separating the real and imaginary components, we obtain the standard real partial derivatives with respect to x:

$$f'(z_0) = \lim_{\Delta x \to 0} \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x} + i \lim_{\Delta x \to 0} \frac{v(x_0 + \Delta x, y_0) - v(x_0, y_0)}{\Delta x} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$$

Second, let h approach 0 along the imaginary axis. We write $h = 0 + i\Delta y$, where $\Delta y \in R$ and $\Delta y \to 0$. The difference quotient evaluates to:

$$f'(z_0) = \lim_{\Delta y \to 0} \frac{u(x_0, y_0 + \Delta y) + iv(x_0, y_0 + \Delta y) - (u(x_0, y_0) + iv(x_0, y_0))}{i\Delta y}$$

Because $\frac{1}{i} = -i$, we factor the imaginary unit to the numerator and separate the components:

$$f'(z_0) = \lim_{\Delta y \to 0} \frac{v(x_0, y_0 + \Delta y) - v(x_0, y_0)}{\Delta y} - i \lim_{\Delta y \to 0} \frac{u(x_0, y_0 + \Delta y) - u(x_0, y_0)}{\Delta y} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

Because f is complex differentiable at z_0, the limit must be strictly path-independent. Equating the real and imaginary parts of the two orthogonal limit evaluations yields:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad and \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

This proves the necessity of the Cauchy-Riemann equations.

[Sufficient Conditions for Holomorphy] Let $f(z) = u(x, y) + iv(x, y)$ be defined on an open set $\Omega \subset C$. If the first-order partial derivatives of u and v exist, are continuous on Ω, and satisfy the Cauchy-Riemann equations, then f is holomorphic on Ω.

The proof of sufficiency relies heavily on the total differentiability of mappings in R^2, leveraging the continuity of the partial derivatives to bound the linear approximation error via the standard Euclidean metric. A profoundly non-trivial consequence of these equations is that any holomorphic function is infinitely differentiable, a fact we will establish via contour integration.

0.1 Cauchy's Integral Theorem, Cauchy's Integral Formula, Morera's Theorem, and Liouville's Theorem

The geometric rigidity of holomorphic functions allows for a profound connection between complex contour integration and the values of a function entirely strictly within a domain.

[Cauchy's Integral Theorem] Let $\Omega \subset C$ be a simply connected open set, and let $f : \Omega \to C$ be a holomorphic function. If γ is a piecewise smooth, simple closed contour contained entirely within Ω, then:

$$\oint_\gamma f(z)\,dz = 0$$

We present the classical proof under the assumption that $f'(z)$ is continuous, which permits the use of Green's Theorem (Goursat later demonstrated that the theorem holds without assuming continuous partial derivatives). Let $f(z) = u(x,y) + iv(x,y)$ and let $dz = dx + idy$. We separate the contour integral into real and imaginary line integrals:

$$\oint_\gamma f(z)\,dz = \oint_\gamma (u+iv)(dx+idy) = \oint_\gamma (u\,dx - v\,dy) + i\oint_\gamma (v\,dx + u\,dy)$$

Because Ω is simply connected and γ is a simple closed contour, γ bounds a simply connected region $D \subset \Omega$. We apply Green's Theorem to both real line integrals:

$$\oint_\gamma (u\,dx - v\,dy) =_D \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dx\,dy$$

$$\oint_\gamma (v\,dx + u\,dy) =_D \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy$$

Because f is holomorphic on Ω, it satisfies the Cauchy-Riemann equations: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. Substituting these into the double integrals yields integrands of identically zero:

$$_D(0)\,dx\,dy + i_D(0)\,dx\,dy = 0$$

Thus, $\oint_\gamma f(z)\,dz = 0$.

[Cauchy's Integral Formula] Let f be holomorphic on a simply connected open set Ω, and let γ be a simple closed contour in Ω oriented counterclockwise. For any point z_0 strictly in the interior of γ:

$$f(z_0) = \frac{1}{2\pi i}\oint_\gamma \frac{f(z)}{z - z_0}\,dz$$

Because Ω is open and z_0 lies in the interior of γ, there exists an $\epsilon > 0$ such that the closed disk $\overline{D_\epsilon(z_0)} = \{z \in C \mid |z - z_0| \leq \epsilon\}$ is entirely contained within the interior of γ. Let C_ϵ denote the boundary circle of this disk, parametrized by $z(\theta) = z_0 + \epsilon e^{i\theta}$ for $0 \leq \theta \leq 2\pi$, oriented counterclockwise.

The function $g(z) = \frac{f(z)}{z-z_0}$ is holomorphic everywhere in Ω except at z_0. By the principle of contour deformation (a corollary of Cauchy's Integral Theorem on multiply connected domains), the integral over γ equals the integral over C_ϵ:

$$\oint_\gamma \frac{f(z)}{z-z_0}\,dz = \oint_{C_\epsilon} \frac{f(z)}{z-z_0}\,dz$$

We rewrite the integrand by adding and subtracting $f(z_0)$:

$$\oint_{C_\epsilon} \frac{f(z)}{z-z_0}\,dz = \oint_{C_\epsilon} \frac{f(z_0)}{z-z_0}\,dz + \oint_{C_\epsilon} \frac{f(z)-f(z_0)}{z-z_0}\,dz$$

We evaluate the first integral by direct parametrization ($dz = i\epsilon e^{i\theta}d\theta$):

$$\oint_{C_\epsilon} \frac{f(z_0)}{z-z_0}\,dz = f(z_0)\int_0^{2\pi} \frac{i\epsilon e^{i\theta}}{\epsilon e^{i\theta}}\,d\theta = f(z_0)\int_0^{2\pi} i\,d\theta = 2\pi i f(z_0)$$

For the second integral, because f is holomorphic, it is continuous at z_0. For any $\eta > 0$, we can choose ϵ sufficiently small such that $|f(z) - f(z_0)| < \eta$ for all $z \in C_\epsilon$. We bound the second integral using the standard estimation lemma (ML-inequality):

$$\left|\oint_{C_\epsilon} \frac{f(z)-f(z_0)}{z-z_0}\,dz\right| \le \max_{z\in C_\epsilon}\left|\frac{f(z)-f(z_0)}{z-z_0}\right| \cdot length(C_\epsilon) < \frac{\eta}{\epsilon}(2\pi\epsilon) = 2\pi\eta$$

Because the original integral over γ is independent of ϵ, and η can be made arbitrarily small by shrinking ϵ, the second integral must be exactly 0. Therefore:

$$\oint_\gamma \frac{f(z)}{z-z_0}\,dz = 2\pi i f(z_0) + 0$$

Dividing by $2\pi i$ yields the result.

[Liouville's Theorem] If $f : C \to C$ is an entire function (holomorphic on all of C) and is bounded, then f is constant.

Because f is bounded, there exists a real number $M > 0$ such that $|f(z)| \le M$ for all $z \in C$. Fix an arbitrary point $z_0 \in C$. By differentiating Cauchy's Integral Formula with respect to z_0, we obtain an integral representation for the derivative:

$$f'(z_0) = \frac{1}{2\pi i}\oint_{C_R} \frac{f(z)}{(z-z_0)^2}\,dz$$

where C_R is a circle of radius $R > 0$ centered at z_0. We apply the estimation lemma to bound the magnitude of the derivative:

$$|f'(z_0)| \le \frac{1}{2\pi}\max_{z\in C_R}\left|\frac{f(z)}{(z-z_0)^2}\right| \cdot length(C_R)$$

Since $|f(z)| \le M$ and $|z - z_0| = R$ on the contour C_R, this yields:

$$|f'(z_0)| \le \frac{1}{2\pi}\frac{M}{R^2}(2\pi R) = \frac{M}{R}$$

Because f is entire, this inequality holds for any arbitrarily large radius R. Taking the limit as $R \to \infty$, we find $|f'(z_0)| \leq 0$, forcing $f'(z_0) = 0$. Since z_0 was chosen arbitrarily, $f'(z) = 0$ for all $z \in C$. A function with a strictly zero derivative on a connected domain is constant.

[Morera's Theorem] Let $\Omega \subset C$ be an open set. If $f : \Omega \to C$ is continuous and satisfies $\oint_\gamma f(z)\, dz = 0$ for every closed piecewise smooth curve γ in Ω, then f is holomorphic on Ω.

Because the integral of f over any closed loop is zero, the integral between any two points is path-independent. Fix $z_0 \in \Omega$ and define a function $F : \Omega \to C$ by:

$$F(z) = \int_{z_0}^{z} f(w)\, dw$$

This function is well-defined due to path independence. We compute the derivative of F directly from the definition:

$$\lim_{h \to 0} \frac{F(z+h) - F(z)}{h} = \lim_{h \to 0} \frac{1}{h} \int_{z}^{z+h} f(w)\, dw$$

Because f is continuous at z, we can write $f(w) = f(z) + (f(w) - f(z))$. Substituting this into the integral:

$$\frac{1}{h} \int_{z}^{z+h} [f(z) + (f(w) - f(z))]\, dw = f(z) + \frac{1}{h} \int_{z}^{z+h} (f(w) - f(z))\, dw$$

As $h \to 0$, w approaches z. By the continuity of f, for any $\epsilon > 0$, there exists $\delta > 0$ such that $|w - z| < \delta |f(w) - f(z)| < \epsilon$. For $|h| < \delta$, the error term is bounded by:

$$\left| \frac{1}{h} \int_{z}^{z+h} (f(w) - f(z))\, dw \right| \leq \frac{1}{|h|} \cdot \epsilon \cdot |h| = \epsilon$$

Thus, the limit evaluates exactly to $f(z)$, proving that $F'(z) = f(z)$. Therefore, F is a holomorphic function on Ω. A profound corollary of Cauchy's Integral Formula is that the derivative of a holomorphic function is itself holomorphic (analytic functions are infinitely differentiable). Since $f = F'$, f must also be holomorphic.

0.1 Power Series, Laurent Series, and the Residue Theorem

While Cauchy's Integral Formula provides an exact representation of a holomorphic function within a simply connected domain, many profound applications of complex analysis involve functions with isolated non-analytic points. To analyze these singularities, we extend the local Taylor series representation to the Laurent series, which accommodates negative powers of the complex variable.

[Laurent's Theorem] Let f be holomorphic in the open annulus $A = \{z \in C \mid r < |z - z_0| < R\}$, where $0 \leq r < R \leq \infty$. Then $f(z)$ can be uniquely represented by a uniformly convergent series on any closed sub-annulus of A:

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

where the coefficients are given by:

$$c_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} \, dw$$

for any simple closed contour γ lying in A and enclosing the inner boundary r, oriented counterclockwise.

The series naturally decomposes into two parts: the analytic part $\sum_{n=0}^{\infty} c_n(z-z_0)^n$, which converges for $|z - z_0| < R$, and the principal part $\sum_{n=1}^{\infty} c_{-n}(z - z_0)^{-n}$, which converges for $|z - z_0| > r$. The behavior of the principal part provides a rigorous classification of isolated singularities.

Let f have an isolated singularity at z_0, and let $\sum_{n=-\infty}^{\infty} c_n(z - z_0)^n$ be its Laurent expansion in the punctured disk $0 < |z - z_0| < R$.

1. If $c_n = 0$ for all $n < 0$, the singularity is **removable**. The function can be extended holomorphically to z_0 by setting $f(z_0) = c_0$.
2. If $c_{-m} \neq 0$ for some integer $m > 0$ and $c_n = 0$ for all $n < -m$, the singularity is a **pole of order** m.
3. If $c_n \neq 0$ for infinitely many negative values of n, the singularity is an **essential singularity**.

The coefficient c_{-1} is of paramount importance. Termed the **residue** of f at z_0 and denoted $Res(f, z_0)$, it is the sole term of the Laurent series that yields a non-zero integral when integrated over a closed loop enclosing z_0, because $\oint_{\gamma} (z - z_0)^n \, dz = 0$ for all integers $n \neq -1$, and equals $2\pi i$ for $n = -1$.

[Cauchy's Residue Theorem] Let $\Omega \subset C$ be a simply connected open set. Let γ be a simple closed contour in Ω, oriented counterclockwise, and let f be holomorphic on Ω except for a finite number of isolated singularities $z_1, z_2, \ldots, z_k$ located strictly in the interior of γ. Then:

$$\oint_{\gamma} f(z) \, dz = 2\pi i \sum_{j=1}^{k} Res(f, z_j)$$

By the principle of contour deformation (generalized Cauchy Integral Theorem for multiply connected domains), we construct small, non-overlapping circles γ_j centered at each singularity z_j, with each γ_j oriented counterclockwise and contained entirely within the interior of γ. The region bounded by γ and the reversed circles $-\gamma_j$ contains no singularities of f. Thus, by Cauchy's Integral Theorem:

$$\oint_\gamma f(z)\,dz + \sum_{j=1}^{k} \oint_{-\gamma_j} f(z)\,dz = 0 \oint_\gamma f(z)\,dz = \sum_{j=1}^{k} \oint_{\gamma_j} f(z)\,dz$$

For each j, the integral over the local circle γ_j isolates the residue at z_j. By expanding $f(z)$ into its Laurent series around z_j, we find:

$$\oint_{\gamma_j} f(z)\,dz = \oint_{\gamma_j} \sum_{n=-\infty}^{\infty} c_{n,j}(z-z_j)^n\,dz = c_{-1,j}(2\pi i) = 2\pi i Res(f, z_j)$$

Substituting this into the contour sum completes the proof.

The Residue Theorem provides a powerful analytic mechanism for evaluating purely real integrals, particularly improper integrals over the entire real line, bypassing the need for explicit antiderivatives.

Application: Evaluating a Real Improper Integral Evaluate the real integral $I = \int_{-\infty}^{\infty} \frac{1}{x^2+a^2}\,dx$ for $a > 0$.

We complexify the integrand, defining $f(z) = \frac{1}{z^2+a^2}$. The function f has isolated singularities where $z^2 + a^2 = 0$, which factors as $(z - ia)(z + ia) = 0$. These are simple poles at $z = ia$ and $z = -ia$. We define a closed semicircular contour $\Gamma = [-R, R] \cup C_R$, where $[-R, R]$ is the segment on the real axis and C_R is the upper semicircle of radius $R > a$ centered at the origin, traversed counterclockwise.

The only singularity of f enclosed by Γ is the simple pole at $z = ia$ in the upper half-plane. We compute the residue at $z = ia$:

$$Res(f, ia) = \lim_{z\to ia}(z - ia)\frac{1}{(z - ia)(z + ia)} = \lim_{z\to ia}\frac{1}{z + ia} = \frac{1}{2ia}$$

By the Residue Theorem, the contour integral evaluates to:

$$\oint_\Gamma f(z)\,dz = 2\pi i Res(f, ia) = 2\pi i \left(\frac{1}{2ia}\right) = \frac{\pi}{a}$$

We decompose the contour integral into its linear and arc components:

$$\oint_\Gamma f(z)\,dz = \int_{-R}^{R} \frac{1}{x^2+a^2}\,dx + \int_{C_R} \frac{1}{z^2+a^2}\,dz$$

We bound the integral over the arc C_R using the estimation lemma (ML-inequality). For $z \in C_R$, $|z| = R$. By the reverse triangle inequality, $|z^2 + a^2| \geq |z|^2 - a^2 = R^2 - a^2$. The length of C_R is πR. Thus:

$$\left|\int_{C_R} \frac{1}{z^2+a^2}\,dz\right| \leq \frac{1}{R^2-a^2}(\pi R)$$

As $R \to \infty$, this bound tends strictly to 0. Taking the limit as $R \to \infty$ of the decomposed contour integral yields:

$$\frac{\pi}{a} = \lim_{R\to\infty} \int_{-R}^{R} \frac{1}{x^2 + a^2}\, dx + 0 = \int_{-\infty}^{\infty} \frac{1}{x^2 + a^2}\, dx$$

Thus, the improper integral converges to $\frac{\pi}{a}$, demonstrating the indispensable utility of analytic continuation and contour deformation in evaluating real-variable phenomena.